Paper 0

AI-Assisted Portfolio Triage at Canada's Infrastructure Crossroads: Integrating Bayesian Scorecards and Cox Survival Models

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Abstract

This paper reviews advancements in AI-assisted portfolio triage, with a focus on integrating Bayesian likelihood ratio scorecards and Cox proportional hazards models for large-scale project evaluation. These approaches address the dual challenges of predicting project outcomes and estimating time-to-event trajectories, enabling both probabilistic classification and survival analysis under uncertainty. Bayesian methods contribute transparent probability estimates and dynamic updating as new data emerge, while Cox models provide flexible time-to-event predictions that accommodate censored data and evolving covariates. Their combined use yields complementary insights into both outcome likelihood and timing, strengthening decision support in infrastructure portfolios. The Canadian context underscores both the urgency and applicability of these methods. Despite committing over $600 billion to natural resource and clean energy projects, Canada faces a deep infrastructure deficit, regulatory uncertainty, and complex Indigenous engagement requirements, while continuing to lag in decarbonization relative to its G7 peers. These challenges amplify the need for predictive, transparent, and adaptable evaluation frameworks. Drawing on applications to Canada's Major Projects Inventory, this review highlights how AI-assisted triage can inform resource allocation, policy evaluation, and risk management, while identifying future research directions and cross-domain applications of probabilistic and survival-based modeling frameworks.

Introduction

Major project evaluation increasingly depends on predictive analytics to manage uncertainty, optimize resource allocation, and guide investment decisions. Infrastructure portfolios such as those catalogued in Canada’s Major Projects Inventory or MPI (Government of Canada 2024b) are characterized by long timelines, diverse risk factors, and high exposure to regulatory, environmental, and market uncertainties (Flyvbjerg 2014). Traditional approaches, including expert judgment, scoring systems, and multi-criteria decision analysis (MCDA), have provided useful structure but are often limited by subjectivity and their inability to model complex interdependencies (Benallou and Aboulaich 2017; Heidenberger and Stummer 1999). Advances in artificial intelligence (AI) and machine learning now enable more rigorous, data-driven approaches to portfolio triage.

Among these, Bayesian likelihood ratio scorecards and Cox proportional hazards models have emerged as complementary tools. Bayesian scorecards provide probabilistic classification of project outcomes, incorporating both prior knowledge and new evidence to generate interpretable, auditable predictions (Shrivastava et al. 2021; Adamantiadou and Tsironis 2025). They are particularly well-suited to dynamic environments where data are incomplete or evolving. By contrast, survival analysis, and especially the Cox model, estimates the timing of events such as project completion or failure, handling censoring and accommodating time-varying covariates (Kvamme et al. 2019). Together, these methods offer both an assessment of “will it build?” and “how much longer?”, a dual perspective crucial for resource prioritization in constrained portfolios.

The convergence of Bayesian and Cox approaches reflects broader methodological trends in predictive analytics. Hybrid models now combine probabilistic reasoning with temporal forecasting, often augmented by ensemble learning, metaheuristic feature selection, or neural networks for pattern recognition (Zéman et al. 2024; Chapfuwa et al. 2020). At the same time, new applications of natural language processing (NLP), topic modeling, and sentiment analysis extend predictive capacity by incorporating unstructured data from project documentation or stakeholder communications (Shafqat et al. 2020).

Despite these advances, persistent challenges remain. Data completeness, quality, and benchmarking vary widely across projects and sectors, limiting transferability and generalizability. Ethical considerations, including bias in historical datasets and the transparency of algorithmic decision support, further complicate adoption (Siemiatycki 2015; Hon et al. 2022). Moreover, in contexts such as Canada, long regulatory timelines and Indigenous consultation obligations underscore the importance of adaptive, interpretable models that can accommodate both technical and socio-political uncertainties (Bishop and Sprague 2019; Wright 2020).

This review synthesizes recent progress in AI-assisted portfolio triage, highlighting methodological innovations, practical applications to Canada's MPI, and emerging opportunities for cross-domain application. By evaluating Bayesian, Cox, and hybrid approaches in tandem, it underscores their potential to transform project selection, policy design, and capital allocation in infrastructure investment under uncertainty.

These methodological debates are not abstract. In Canada, major project evaluation occurs against the backdrop of unprecedented investment commitments, persistent infrastructure deficits, and complex regulatory and socio-political dynamics. Understanding this national context is essential for situating AI-assisted portfolio triage within real-world policy and market conditions.

Canada's Infrastructure Crossroads

Canada is proposing substantial national investments in major projects, with over $632.6 billion allocated to natural resource initiatives, encompassing energy and critical minerals, and supported by programs like the Canada Infrastructure Bank, Canada Growth Fund and Clean Economy Investment Tax Credits (Government of Canada 2025; Canada Growth Fund, n.d.; Government of Canada 2002; 2024c; 2024b). Yet, the nation faces a "deep infrastructure deficit," estimated at up to $600 billion, highlighting an urgent need for critical infrastructure development (Khanal et al. 2023).

Project advancement is significantly hindered by regulatory uncertainty, as processes are slower and less predictable than international competitors, often due to broad legislation like the Impact Assessment Act that invites litigation and has led to quashed federal approvals from insufficient Indigenous consultation, with Canadian environmental assessments taking longer than those in Australia (Bishop and Sprague 2019; DeLand and Gilmour 2024). Indigenous engagement presents complex legal and historical challenges, resulting in high transaction costs and delays, as meaningful consultation is difficult to achieve consistently across diverse communities, though efforts include fostering Indigenous-led institutions (Le Dressay et al. 2022; Wright 2020). Project performance is frequently impacted by cost overruns and schedule delays, largely stemming from poor planning and scope definition rather than labor productivity, and Public-Private Partnerships (P3s) show substantial budget underestimation (Chanmeka et al. 2012;

Zhong et al. 2022). Furthermore, Canada's investment climate struggles to attract foreign direct investment and private capital due to policy instability, and the country lags G7 nations in emission reduction, with increased critical mineral mining posing new environmental risks (Khanal et al. 2023; Bishop and Sprague 2019; Government of Canada 2024d; Kaiser et al. 2024). Opportunities for improvement exist in streamlining approvals, strengthening Indigenous partnerships, and enhancing project planning and financing, drawing lessons from programs like the Investing in Canada Infrastructure Program (Government of Canada 2024a).

Foundations of Portfolio Risk Assessment

Classical approaches to portfolio risk assessment relied heavily on expert judgement and scoring models. These methods translated qualitative assessments of feasibility, cost, and strategic fit into structured frameworks, often through weighted checklists or additive scoring systems (Jikiemi 2024; Benallou and Aboulaich 2017). While useful for imposing consistency, they were limited by subjectivity, vulnerability to cognitive bias, and weak reproducibility. Multi-criteria decision analysis (MCDA) advanced this tradition by incorporating probabilistic simulation and structured trade-offs among cost, risk, and strategic alignment. Tools such as Monte Carlo simulation enabled more realistic treatment of uncertainty but remained computationally intensive and often detached from real-time project dynamics (Heidenberger and Stummer 1999; Khodabakhshian et al. 2023).

The introduction of statistical methods brought greater rigor. Linear regression supported cost and schedule forecasting, while logistic regression enabled binary predictions of project success or failure (J. Zhang et al. 2019). Decision trees and ensemble methods such as random forests and gradient boosting later improved flexibility by modeling nonlinear relationships, handling mixed data types, and providing transparent variable importance rankings (Adamantiadou and Tsironis 2025). Ensembles also reduced variance and mitigated overfitting, making them attractive for large heterogeneous datasets such as Canada's MPI (Hon et al. 2022).

Despite these gains, classical and machine-learning approaches were constrained by their siloed perspectives: scoring models classified outcomes but ignored timing, while regression or tree-based methods predicted performance without quantifying uncertainty. These limitations motivated the development of blended frameworks that integrate probabilistic classification with survival analysis to capture both outcome likelihood and time-to-event dynamics.

Bayesian Likelihood Ratio Scorecards

Bayesian methods have become central to AI-assisted portfolio triage because of their ability to quantify uncertainty and dynamically update predictions. The Bayesian paradigm combines prior knowledge with new evidence, producing posterior probabilities that reflect both historical data and current observations (Shrivastava et al. 2021; Z. Zhang et al. 2022). In project evaluation, this supports interpretable predictions of success or failure that can adapt as conditions evolve—an essential capability for large-scale infrastructure portfolios subject to regulatory shifts, market volatility, and stakeholder pressures.

Likelihood ratio scorecards operationalize this approach by assigning weights to project features based on how strongly they distinguish successful from unsuccessful outcomes. Unlike deterministic scoring systems, Bayesian scorecards express these weights probabilistically, allowing outputs to be interpreted directly as measures of confidence (Ning and Castillo 2024). For example, factors such as project size, sector, or proponent experience can be converted into likelihood ratios, which are then aggregated to produce a probability of success. This probabilistic framework enhances transparency: each variable's contribution is explicit and auditable, a feature valued in high-stakes environments where accountability is critical (Zéman et al. 2024).

Recent advances extend Bayesian models to survival contexts, enabling them to estimate not only outcome probabilities but also event timing when combined with hazard models (Chapfuwa et al. 2020). Bayesian survival analysis has shown higher classification accuracy than frequentist Cox models in some settings, particularly when data are incomplete or noisy (Shrivastava et al. 2021). Nonparametric Bayesian adjustments further address unmeasured confounders by clustering projects with similar hidden characteristics, improving robustness in heterogeneous datasets such as Canada's MPI (Orihara 2023).

Interpretability remains a defining strength. Posterior distributions and credible intervals communicate both estimates and their uncertainty, avoiding the false precision of point forecasts (Kuss and Hoyer 2021). This makes Bayesian models particularly useful for policy-facing applications, where decision-makers require transparent explanations of risk. At the same time, methodological challenges persist: specifying appropriate priors, ensuring calibration under partial observability, and managing computational complexity when dealing with high-dimensional data (Lázaro et al. 2020).

In practice, Bayesian scorecards now serve as both early-warning systems and resource-allocation tools. By ranking projects according to their probability of success and updating dynamically as new data arrive, they allow portfolio managers to triage under uncertainty. When integrated with complementary models such as Cox regression, they form the foundation of blended frameworks capable of addressing both outcome classification and time-to-event forecasting in infrastructure investments.

Cox Proportional Hazards Models

Survival analysis provides a framework for estimating not only whether a project will succeed or fail but also when such events are likely to occur. Among survival methods, the Cox proportional hazards (Cox PH) model has become the most widely applied in project evaluation because it balances flexibility with interpretability. The Cox model estimates the hazard rate—the instantaneous probability of event occurrence at a given time—without requiring specification of the baseline hazard, making it semi-parametric and adaptable to heterogeneous project portfolios (Simon et al. 2025; Kvamme et al. 2019).

The Cox model's strength lies in its ability to handle censored data, which are common in infrastructure portfolios where many projects are ongoing or unfinished at the time of analysis. By incorporating censored observations, it leverages all available information without discarding partially observed cases. The model also accommodates time-varying covariates, allowing risk profiles to evolve as projects progress, for example, when new funding is secured, permits are issued, or costs escalate (Moussa et al. 2024).

Interpretation is another advantage. Coefficients are expressed as hazard ratios, which quantify the multiplicative effect of covariates on event risk. This provides actionable insights: a hazard ratio greater than one indicates increased risk of project termination or delay, while values below one suggest protective factors. Such outputs are intuitive for practitioners and policymakers tasked with prioritizing interventions. However, the model depends on the proportional hazards assumption that relative risks remain constant over time, which may not hold in complex projects with shifting risk dynamics (Kuss and Hoyer 2021). Violations can reduce predictive accuracy unless addressed through stratification, time-dependent effects, or neural network extensions such as DeepSurv (Kvamme et al. 2019).

Comparative studies show that while traditional Cox models achieve acceptable discrimination (concordance indices above 0.7 in some cases), enhanced variants incorporating non-linear covariates, Bayesian extensions, or ensemble methods often improve calibration and forecasting performance (Chapfuwa et al. 2020; Lázaro et al. 2020). Bayesian Cox models, in particular, provide posterior distributions for hazard ratios, enabling full uncertainty quantification and more robust inference under sparse or noisy data (Orihara 2023).

In the context of portfolio triage, Cox models are most valuable when combined with probabilistic classifiers. While Bayesian scorecards predict "if," Cox models predict "when," giving decision-makers both a probability surface and a temporal dimension. This dual perspective is particularly important in managing Canada's MPI dataset, where understanding both likelihood and timing of project milestones is essential for resource allocation, scheduling, and policy evaluation.

Hybrid and AI-Driven Extensions

While Bayesian scorecards and Cox models provide complementary strengths, recent advances increasingly focus on hybrid frameworks that integrate probabilistic reasoning with machine learning and deep learning techniques. These blended approaches aim to capture both outcome likelihoods and event timing while improving accuracy, scalability, and interpretability in complex datasets.

Hybrid Models. Ensemble and hybrid models combine statistical foundations with machine learning flexibility. For example, Bayesian scorecards can be coupled with Cox survival models to jointly estimate success probabilities and time-to-event outcomes. Machine learning algorithms such as random forests or gradient boosting improve feature selection, handle nonlinearities, and reduce overfitting in high-dimensional datasets typical of national project inventories (Alaka et al. 2018; Zéman et al. 2024). Metaheuristic optimization techniques, including genetic algorithms and particle swarm optimization, further enhance variable selection and calibration, ensuring predictive stability across diverse project types (Erwin and Engelbrecht 2023).

Deep Learning. Neural architectures extend survival and classification models by learning complex interactions between covariates and time. Long Short-Term Memory (LSTM) networks capture sequential dependencies in project lifecycles, modeling how early signals such as funding delays or regulatory approvals propagate through time (Shafqat et al. 2020). When combined with topic modeling, LSTMs can align semantic content from project documents with temporal dynamics, enriching early-warning systems. Similarly, convolutional neural networks (CNNs) have been applied with metaheuristic-based feature selection to improve predictions of crowdfunding or infrastructure outcomes, demonstrating broader cross-domain potential (Yab et al. 2022).

Natural Language Processing (NLP). Much critical project information exists in unstructured text, such as regulatory filings, environmental assessments, or stakeholder submissions. Topic modeling methods such as Latent Dirichlet Allocation (LDA) uncover thematic structures in project documentation, while sentiment analysis quantifies stakeholder confidence or skepticism. Integrating these signals with structured covariates enhances predictive accuracy and contextual relevance (Adamantiadou and Tsironis 2025; Sharma et al. 2022). For instance, sentiment mining of investor commentary has been shown to correlate strongly with project credibility, offering a soft indicator that complements quantitative risk scores (Shafqat et al. 2020).

Handling Data Quality. Hybrid frameworks also tackle pervasive challenges of missing, noisy, or censored data. Bayesian methods address incomplete histories by averaging across unobserved outcomes, while machine learning imputation and feature selection reduce variance and bias (Lin and Yuan 2019). Survival forests and neural survival models relax the proportional hazards assumption, improving calibration in heterogeneous project datasets (Kvamme et al. 2019).

Model Evaluation. Validation remains critical to adoption. Discrimination metrics such as the concordance index (C-index) assess survival predictions, while calibration plots and Brier scores test probabilistic accuracy. Proper scoring rules like the Continuous Ranked Probability Score (CRPS) provide comprehensive evaluation across thresholds, particularly for hybrid Bayesian–survival frameworks (Chapfuwa et al. 2020). Beyond statistical metrics, expert validation and sensitivity analysis ensure that critical success factors, such as managerial competence or stakeholder engagement, remain reflected in model outputs (Yamany et al. 2024).

Taken together, these innovations mark a shift from siloed statistical tools toward blended AI systems that integrate structured and unstructured data, probabilistic reasoning, and temporal modeling. By combining interpretability with predictive power, hybrid and deep learning models extend the utility of Bayesian and Cox approaches, offering more comprehensive decision support for large-scale infrastructure portfolios.

Applications to Canada's Major Projects Inventory

The Canada Major Projects Inventory (MPI) provides a unique testbed for AI-assisted portfolio triage. Compiled annually by Natural Resources Canada, the MPI tracks more than 500 large-scale projects across energy, mining, and forestry, including variables such as sector, geography, cost estimates, status updates, and planned timelines. Its breadth and longitudinal coverage make it well suited for survival and probabilistic modeling (Khanal et al. 2023).

Applied to this dataset, Bayesian likelihood ratio scorecards offer transparent probability estimates of project advancement, classifying projects into success and failure trajectories based on features such as sector, ownership, and capital cost. These predictions are particularly valuable given the MPI's heterogeneous data quality: Bayesian updating accommodates missing or evolving information, making outputs robust even under partial observability (Adamantiadou and Tsironis 2025).

Cox proportional hazards models extend this analysis by estimating time-to-event outcomes, such as the duration to construction start or likelihood of delay. The semi-parametric nature of the Cox model is advantageous for censored cases, since many MPI projects remain ongoing or incomplete at the time of reporting (Kvamme et al. 2019). By incorporating time-varying covariates, such as shifts in regulatory status or financing milestones, Cox models enable dynamic risk profiling and allow policymakers to anticipate bottlenecks.

Hybrid frameworks that blend Bayesian classifiers with survival models yield a richer risk surface. For example, a project may have a high probability of eventual success but a long-expected time-to-construction, suggesting different resource allocation strategies than a project with imminent but uncertain prospects. Integrating structured MPI fields with unstructured documentation, such as environmental assessments or stakeholder submissions, further enriches prediction accuracy (Shafqat et al. 2020).

In this way, the MPI illustrates both the promise and the challenges of AI-assisted triage. It provides the structured baseline needed for quantitative modeling but also highlights issues of data completeness, standardization, and cross-sector comparability that remain barriers to scalable adoption.

Challenges: Data, Uncertainty, and Ethics

Despite significant methodological advances, AI-assisted portfolio triage faces persistent challenges that limit its real-world adoption. Chief among these are issues of data quality, uncertainty, and ethics.

Data quality and completeness. Infrastructure datasets like Canada's MPI often contain missing values, outdated entries, or inconsistencies across sectors and jurisdictions. Such gaps undermine both Bayesian and Cox models: incomplete event histories bias hazard estimates, while sparse categorical fields reduce the reliability of probabilistic classifiers (Lin and Yuan 2019). Although imputation techniques and Bayesian averaging can mitigate some effects, robust predictions require systematic collection and validation of project performance data, which remains underdeveloped in many jurisdictions (Siemiatycki 2015).

Uncertainty and bias. AI models risk amplifying subjectivity and historical inequities. For example, expert-driven inputs or survey-based fields may embed cognitive biases, while historical records can reflect structural disadvantages in funding or approval processes (Afful-Dadzie and Afful-Dadzie 2016). Bayesian frameworks explicitly quantify parameter uncertainty through posterior distributions, yet their accuracy is still constrained by the representativeness of input data (Orihara 2023). Survival models, meanwhile, rely on proportional hazard assumptions that may not hold in dynamic, politically sensitive project environments (Kuss and Hoyer 2021; Hartman et al. 2023).

Ethical and governance concerns. As AI-driven predictions increasingly inform high-stakes investment and policy decisions, transparency and fairness become essential. Opaque models risk eroding trust if stakeholders cannot trace how inputs influence outputs. Bayesian scorecards and Cox regressions are relatively interpretable, but deep learning extensions pose challenges for explainability (Hon et al. 2022). Moreover, algorithmic triage may inadvertently reinforce systemic inequities unless fairness audits, subgroup validations, and ethical oversight are embedded in deployment frameworks.

Addressing these challenges requires more than technical innovation. Improved data governance, cross-industry benchmarking, and stakeholder engagement are necessary to ensure that AI-assisted decision systems deliver both predictive accuracy and equitable outcomes in project portfolio management.

Future Directions and Cross-Domain Potential

The integration of Bayesian scorecards and Cox survival models into hybrid AI frameworks points toward several future research directions. Methodological innovation is likely to focus on relaxing proportional hazard assumptions, improving calibration under censoring, and combining probabilistic classification with sequence-based deep learning models such as LSTMs for temporal forecasting (Kvamme et al. 2019; Shafqat et al. 2020). Advances in feature selection through metaheuristic optimization and explainable AI will also be critical to enhance interpretability while managing high-dimensional data (Zéman et al. 2024).

Beyond infrastructure, these methods hold cross-domain potential wherever outcome prediction and time-to-event estimation are central. In healthcare, Bayesian–Cox frameworks are already applied to survival analysis in oncology; in finance, they can predict loan defaults or bankruptcy risk; in engineering, they support equipment failure forecasting. Their adaptability stems from shared strengths: transparent uncertainty quantification, capacity to handle censored or incomplete data, and ability to combine structured and unstructured sources (Kraisangka and Druzdzel 2014; Yeh and Chen 2020).

For infrastructure policy, the next frontier lies in integrating qualitative credibility assessments, such as stakeholder sentiment and regulatory narratives, into structured portfolio analytics. Doing so would move AI-assisted triage beyond technical risk prediction toward holistic decision support, balancing economic, social, and environmental priorities.

Conclusion

AI-assisted portfolio triage represents a significant advance in project evaluation, particularly when Bayesian likelihood ratio scorecards and Cox proportional hazards models are integrated into blended frameworks. Together, these approaches provide complementary insights into both the probability of project success and the timing of critical events, enabling more informed decisions in resource allocation and risk management. Hybrid extensions that draw on deep learning, NLP, and metaheuristic optimization will further enhance predictive performance, though they raise new challenges of interpretability, data quality, and fairness.

In Canada, the stakes are especially high. The country faces a deep infrastructure deficit even as it commits over $600 billion to major projects in natural resources and clean energy. Yet, regulatory uncertainty, litigation risk under broad legislation, and the complexity of Indigenous engagement continue to delay project advancement, while cost overruns and weak planning undermine delivery. At the same time, Canada must balance attracting investment with meeting ambitious decarbonization goals. In this context, AI-assisted triage offers a pathway to greater transparency, accountability, and predictive rigor, helping policymakers and investors prioritize scarce resources and accelerate the transition toward sustainable, resilient infrastructure. Canada's Major Projects Inventory provides a uniquely comprehensive dataset where these methods can be piloted, tested, and refined, linking methodological innovation directly to the country's infrastructure crossroads.

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